

Biruk Ambaye

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Summary

Embedded Systems Master's student at Uppsala University with a background in low-level firmware development and RTOS-based sensing systems. Experienced in developing Zephyr RTOS applications for ARM (Cortex-M33) and building deterministic schedulers for AVR architectures. Currently conducting research on dyadic backscatter communication, investigating baud rate optimisation to improve reliability in passive RF systems using 2-FSK modulation (NRF52840, Raspberry Pi Pico, CC1352). Proficient in I2C data acquisition, real-time TPHG monitoring, and sub-GHz wireless protocols, with a consistent focus on software quality and agile development standards.

Education

Uppsala University — MSc in Embedded Systems 2024 – Present
Specialization in Real-Time Systems and Microcontroller Interfacing Sweden

Mekelle University — BSc in Computer Science 2017 – 2024
Ethiopia

Technical Skills

Languages: C, C++, Python, JavaScript (React.js)

Embedded: Zephyr RTOS, ARM (Cortex-M33/RP2350), AVR (Atmega), FreeRTOS, Raspberry Pi Pico (RP2040/RP2350)

Wireless & RF: Dyadic backscatter, 2-FSK modulation, sub-GHz communication (NRF52840, CC1352, CC2500), sync-word framing, BER/PER analysis

Connectivity: I2C, UART, SPI, CAN bus, USB serial

Tools: Logic Analyzers, Oscilloscopes, GDB Debugging, Git, Linux/POSIX, SmartRF Studio, Python data analysis

Embedded Projects

Dyadic Backscatter Communication — Baud Rate Optimisation Research
Research Project / Uppsala University (WCNES course) / NRF52840, Raspberry Pi Pico, CC1352, CC2500 2026

- Investigated baud rate reduction (100 kBaud → 50 kBaud) as a technique to improve SNR and reliability in a passive backscatter channel, without increasing transmit power.
- Implemented a complete dyadic backscatter testbed: NRF52840 as CW carrier, Raspberry Pi Pico with two CC2500 antennas as the passive 2-FSK tag, and CC1352 as the receiver.
- Collected and analysed 15 measurement logs (5 carrier distances × 3 baud rates) comparing Bit Error Rate (BER) and Packet Error Rate (PER).
- Wrote a custom Python analysis script handling multiple SmartRF Studio log formats with automatic format detection, computing BER and PER statistics per configuration.
- Validated CC1352 sync-word detection behaviour by deliberately injecting single-bit errors, confirming hardware-level packet rejection.

Health Monitoring System

Group Project / Zephyr RTOS, Raspberry Pi Pico 2, KY-039, MIKROE-3102 SpO2, BME680, ADC, I2C, UART 2026

- Built a two-node health monitoring device running Zephyr RTOS: a sensor node collecting heart rate (KY-039 via ADC), blood oxygen (MIKROE-3102 via I2C), and body temperature (BME680 via I2C).
- Implemented UART communication between the sensor node and a base station for data processing and real-time USB serial visualisation on a host computer.

Thermal Safety and Hardware Alert Prototype

Zephyr RTOS / RP2350 (Cortex-M33) 2026

- Engineered a real-time hardware alert system using software-defined PWM to drive a passive buzzer for high-temperature notifications.

- Utilised Devicetree Overlays (**.overlay**) to define the buzzer node and Kconfig (**.conf**) to enable PWM and Sensor drivers within the Zephyr kernel.
- Developed the application using the Zephyr Sensor API to ensure thread-safe data acquisition and modular hardware abstraction.

Environmental Monitoring & Air Quality System

Zephyr RTOS / RP2350 (Cortex-M33)

2026

- Developed firmware to control a BME680 sensor, managing the integrated heater plate for Metal-Oxide (MOX) gas sensing and VOC detection.
- Implemented real-time monitoring of Gas Resistance by heating the sensing element to specific target temperatures to detect Volatile Organic Compounds (VOCs).
- Processed multi-modal TPHG data (Temperature, Pressure, Humidity, Gas) using the Zephyr Sensor API to calculate comprehensive Indoor Air Quality (IAQ) scores.

Real-Time Scheduler

Embedded C / AVR (Atmega)

2025

- Built a deterministic scheduler for AVR microcontrollers to manage periodic task execution.
- Performed WCET analysis to guarantee hard real-time deadline compliance.

Professional Experience

Aynalem Electronics — Junior Frontend Web Developer

2024 – 2025

- Developed modular and reusable React.js interfaces, applying strict software quality and code review standards to ensure maintainable codebases.
- Optimized front-end performance and state management logic, ensuring seamless user experiences across diverse web platforms.

Proxima Technologies — Frontend Web Developer

2023 – 2024

- Utilised Git version control within an agile Scrum environment to collaborate on high-velocity feature delivery and documentation.
- Collaborated in cross-functional teams to translate complex requirements into functional, high-performance software features.

Languages

English: Professional working proficiency **Swedish:** Basic **Amharic / Tigrigna:** Native